Which Optimizers Find the Support? An Empirical Comparison on Sparse-Support Tasks

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***Abstract*—**

I. Introduction

A trained neural network must identify the *support* of its target function – the subset of input coordinates that actually influence the output – to learn an interpretable and robust representation. Beneventano et al. [1] show that this identification typically happens in the *first layer*, but only under specific optimization regimes: full-batch gradient descent (GD) approximates the target well, yet its first-layer weights interacting with irrelevant input coordinates do not converge to zero. Mini-batch SGD, in contrast, drives those weights to zero, and does so faster as the ratio η/b grows. The paper attributes this to a second-order implicit regularization term that arises from the discrepancy between the mini-batch and full-batch input covariance matrices.

The original study compares GD, SGD at three batch sizes, and GD with weight decay. Several optimizers that are standard in modern deep learning – Adam, AdamW, and the recent Muon optimizer [2] – are not studied. They differ from SGD in important ways: Adam uses coordinate-wise adaptive learning rates, AdamW decouples weight decay from the gradient update, and Muon orthogonalizes momentum updates via Newton–Schulz iterations. Whether the support-finding mechanism described by [1] survives, disappears, or is replaced by a different mechanism under these optimizers is, to the best of our knowledge, an open empirical question.

Contributions. (i) We reproduce the GD vs. SGD vs. GD+wd comparison of [1] on a sparse-support regression problem. (ii) We extend the comparison to Adam, AdamW, and Muon on three target functions (linear, sine, staircase). (iii) We replicate the CIFAR10 transfer-learning setup of the paper (frozen pretrained ResNet + trainable MLP head) under the same optimizer set, and compare support-suppression in the first layer of the MLP head. (iv) We summarize each optimizer with three diagnostics: final loss, norm of W_1 restricted to irrelevant coordinates, effective rank of W_1, stable rank [TODO].

II. Related Work

Implicit regularization of SGD. The fact that mini-batch SGD selects different minima from GD is well documented. Keskar et al. [3] link large-batch training to sharper minima and worse generalization. Goyal et al. [4], Jastrzebski et al. [5] and He et al. [6] formalize the *linear scaling rule* between learning rate and batch size, suggesting that the implicit regularization strength scales as η/b. Smith et al. [7] and Beneventano [8] derive modified-loss interpretations of SGD without replacement.

Support learning and feature selection in neural networks. A line of work studies how neural networks discover a low-dimensional support in the infinite-data regime, often on parity-like or single-index targets [9], [10]. Closest to our setting, Beneventano et al. [1] show that on a finite dataset, support identification in the *first layer* is an artifact of the optimizer, not of the target function itself.

Adam, AdamW, Muon. Coordinate-wise adaptive methods have been argued both to dampen and to amplify the implicit bias of plain SGD [11], [12]. Muon [2] replaces the gradient update on matrix parameters with an approximately orthogonal version, an explicit deviation from standard first-order methods. We are not aware of prior work specifically comparing these optimizers on the support-identification task.

III. Experimental Setup

A. Sparse-support synthetic data

Following [1], we generate a dataset $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^{2n}$ with $n = 5000$, input dimension $d = 15$, and $r = 5$ *relevant* coordinates. Inputs $x \in \mathbb{R}^d$ are drawn from $\mathcal{N}(0, I_d)$, and the irrelevant coordinates $x[r\!:]$ are centred to remove any spurious covariance with the relevant ones. The target depends only on $x[\,:\!r]$:

$$y(x) = \begin{cases} \sum_{i=1}^{r} x_i & \text{(linear)}, \\ \sin(\sum_{i=1}^{r} x_i) & \text{(sine)}, \\ \sum_{i=1}^{r} x_i^i & \text{(staircase)}, \end{cases} \tag{1}$$

To match the protocol of [1] and induce label oscillations, each x_i is duplicated with labels $y_i \pm \varepsilon$, where $\varepsilon \sim \mathcal{N}(0, \sigma^2)$ and $\sigma = 10^{-3}$. This symmetric $\pm\varepsilon$ pair forces residuals to oscillate around zero even at a perfect linear fit, which is what powers the second-order SGD effect of [1].

B. Model

We use a fully connected MLP with three hidden layers of width 15 and a scalar output, without biases. This matches the linear-network setting in which Theorem 1 of [1] is stated. Weights are initialised with He-normal. All experiments use the MSE loss. We report results for the *linear* activation (closest to the theory) and for ReLU (closest to practice).

C. Optimizers compared

We compare seven optimizer configurations, listed in Table **??**. The first three (GD, SGD at two batch sizes, GD+wd) reproduce the comparison of [1]. The last three (Adam, AdamW, Muon) extend it. All learning rates and weight-decay values are tuned per target on a held-out validation fraction; the final values used are reported in Table **?? [TODO: fill]**.

D. Metrics

For each run we record, at every gradient step: the training loss, and the Frobenius norm of the irrelevant block $\|W_1[:, r{:}d]\|_F$ of the first-layer weight matrix. After training we additionally compute, per layer: the Gram matrix $W^\top W$, its eigenvalue spectrum, and the effective rank $\exp\left(-\sum_i p_i \log p_i\right)$ with $p_i = s_i / \sum_j s_j$ over singular values s_i.

E. CIFAR10 setup

[TODO]

IV. Results

A. Reproducing GD vs. SGD

[TODO Fig. ??: norm of irrelevant first-layer weights over time, log-scaled x-axis, one curve per optimizer, linear target.] We first verify that our pipeline reproduces the main finding of [1]: on the linear target, all optimizers reach comparable training loss, but only SGD and GD+wd drive $\|W_1[:, r{:}d]\|_F$ to zero. GD plateaus at a non-zero level. The ranking of SGD-32 vs. SGD-512 is consistent with the η/b scaling.

B. Adam

[TODO Fig. ??: same plot, with Adam/AdamW/Muon added.] The question is whether the implicit-regularization story carries over. **[TODO: 2-3 sentences with the observed pattern – e.g. whether AdamW behaves like GD+wd, whether Muon's orthogonalization helps or hurts, whether Adam's per-coordinate scaling masks the η/b effect.]**

C. Muon

D. CIFAR10

[TODO Fig. ??: eigenvalue histogram of $W_1^\top W_1$ for the MLP head, one subplot per optimizer, with test accuracy in the title.] We replicate the qualitative finding of Appendix D of [1]: at matched test accuracy, optimizers differ markedly in how compactly W_1 uses the 512-dimensional ResNet feature space.

V. Discussion

[TODO: a paragraph summarising which optimizers behave like SGD and which behave like GD on this task. Likely structure: (i) AdamW vs Adam, (ii) Muon, (iii) practical implication for interpretability of W_1.]

Limitations. Our experiments are restricted to MLPs and to a transfer-learning setting on CIFAR10. The synthetic targets are designed so that "irrelevant" is unambiguous; for general non-linear targets, the notion of irrelevance becomes more subtle (Section 6 of [1]). We do not retune optimizer-specific hyperparameters at the scale a production pipeline would, and Muon in particular is known to be sensitive to its momentum and Newton–Schulz parameters.

VI. Conclusion

[TODO: 2-3 sentences once results are in.] Our experiments **[confirm/refine/contradict]** the picture of [1]: implicit support identification in the first layer is a property of **[which optimizers]**, and **[is/is not]** an automatic consequence of using stochastic or adaptive updates.

Reproducibility

All code, configuration files, and plotting notebooks are available at **[TODO: GitHub URL]**. The script `train_baseline.py` reproduces every run reported in this paper; `visualization.ipynb` regenerates every figure.

References

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